Prodigy Anesthesia Quick Review

Electrocardiogram

Which exerts greater control over cardiac contractility, the sympathetic or parasympathetic nervous system?	The sympathetic nervous system exerts greater control. It can increase contractility by about 100% whereas the parasympathetic can decrease contractility by only about 30%. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
what are the first two vessels to branch off of the aorta?	The right and left coronary vessels *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the velocity of conduction within the Purkinje network and why is this important?	2-4 meters per second. The very fast transmission allows for uniform ventricular contraction. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the velocity of electrical conduction within the proximal portion of the AV node? Why is this region important?	0.02 meters per second. The proximal AV node consists of junctional fibers which are responsible for 1/2 of the delay that occurs in the AV node. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
Which portion of the cardiac conduction system generates fast response action potentials?	Atrial and ventricular myocytes and the His-Purkinje cells. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
Contrast the methods by which calcium channel blockers and lidocaine affect cardiac function.	Calcium channel blockers extend refractoriness in pacemaker cells (slow-response) and lidocaine decreases the slope of phase 4 (automaticity) in ischemic His-Purkinje cells (fast-response). *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What components of the cardiac conduction system generate slow response action potentials?	SA node, internodal tracts, and the bundle of His *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the significance of a short PR Interval?	It can indicate Wolff-Parkinson-White syndrome if associated with a delta wave. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
Does the parasympathetic or sympathetic nervous system exert greater control over the heart rate?	Parasympathetic *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*

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What is a biphasic P wave and what is its significance?	A P wave that moves above the isoelectric line and then dips below it before returning to the isoelectric line. It can indicate atrial hypertrophy. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the incidence and significance of complete heart block in anterior and inferior MIs?	The incidence of CHB is higher in inferior infarcts (11%) than anterior (5%), but the potential for poor outcome is much more significant in anterior MIs. CHB following anterior MI is associated with a 81% mortality rate versus 36% for inferior MIs. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What areas of the heart does the circumflex supply?	It gives off branches to the left lateral ventricular wall, the sino-atrial node (in 45% of the population) and the AV node (in 10% of the population). *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What types of arrythmias would you be likely to see following an inferior or posterior MI?	Obstruction of the RCA or circumflex typically results in SA and AV nodal dysrhythmias such as sinus arrest, sinus pause, type I AV block, nodal rhythms, and complete heart block. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What dysrhythmias would you be likely to see with an anterior MI?	Wide complex rhythms such as bundle branch block, complete heart block, mobitz type II block, and idioventricular rhythms. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
Describe the anatomy of the PDCA with respect to variations among the population.	In approximately 2-30% of the population the PDCA is derived from the circumflex (left-dominant) and in 50-95% of the population it is derived from the RCA. In about 3-20% of the population, no true PDCA exists. In this situation the posterior and inferior aspects of the heart are supplied by a network of branches arising off the RCA and circumflex. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
The PDCA is right dominant in what percentage of the population?	50-95% *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
How would you expect a normal R wave to change as you move from V1 to V6?	It should become progressively larger when moving from V1 to V6. It typically is largest, however at V5. From V1 the deflection should be almost entirely negative, it should be equal at V3-V4 and should be completely positive at V6. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*

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What center in the brain effects a balance between the sympathetic and the parasympathetic systems to control heart rate and contractility?	The vasomotor center in the lower third of the pons and medulla. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What degree of ST segment deviations are indicative of myocardial injury?	Deviations greater than or equal to 1mm *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
How do you determine the normal QT interval?	For a heart rate of 60-70, it shouldn't be longer than 0.4 secs. Outside this range it must be corrected using Bazett's formula: QTc=QT/square root of the R-R interval *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What are possible causes of U waves with negative deflections?	Proximal LAD disease, myocardial ischemia, increased LV afterload, aortic or mitral regurgitation. Sometimes there is no apparent reason for the presence of U waves. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is clockwise and counterclockwise rotation with respect to the V leads?	Clockwise rotation means that the transitional leads (equally positive and negative deflections) shift to the left and counterclockwise shifts to the right. For example, normally the V3-V4 leads should have as much deflection above the isoelectric line as above it. In a clockwise rotation, the V5-V6 leads become the transitional leads and in counterclockwise rotation the V1-V2 leads become transitional. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
In a normal 3-lead ECG setup, where do you place the positive and negative electrodes for leads I, II, and III?	In lead I, the right arm is negative and the left arm is positive. In lead II, the right arm is negative and the left leg is positive. In lead III, the left arm is negative and the left leg is positive. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is a U Wave and what is it's significance?	It is a small positive wave that follows the T wave. It can be associated with hypokalemia, hypercalcemia, quinidine, digitalis, epinephrine, intracranial hemorrhage, or papillary muscle dysfunction. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What are possible causes of a prolonged QT interval?	1. CHF 2. Ischemic heart disease 3. Prinzmetal's angina 4. Rheumatic fever 5. Cerebrovascular disease 6. Subarchanoid hemorrhage 7. Hypocalcemia 8. Procainamide 9. Disopyramide 10. Phenothiazines 11. Amiodarone 12. Hypothermia 13. Mitral valve prolapse *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*

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What is the speed in mm/sec that EKG paper moves past the stylus under standard settings?	25mm/sec *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
How do you measure the QT interval?	From the beginning of the Q wave to the end of the T wave. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What two vessels does the LCA branch into?	The LAD and the circumflex *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the definition of the S wave?	The first negative deflection following the R Wave (you can't have an S wave without an R wave). *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the definition of the R Wave?	It is the first positive deflection of ventricular depolarization. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
Where would you expect to best see U waves if present?	V3 *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the normal PR Interval?	0.12-0.2 seconds *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the definition of the Q wave?	The first negative deflection after the P wave. It begins as it leaves the isoelectric line and terminates as it returns to the isoelectric line. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the definition of the isoelectric line on an ECG?	It is the baseline of the ECG where no electrical activity is taking place. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What voltage does 10 mm in height correspond to in mV on ECG paper and how can you verify this?	It should be 1mV. You can verify this by viewing the calibration pulse of 1 mV that appears before the ECG tracing to make sure it equals 10 small squares (10mm) in height. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*

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The PR interval is normally isoelectric. What can cause it to deviate from the isoelectric line?	Atrial infarct or pericarditis. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What three structures drain blood into the left atrium?	1. Left pulmonary veins 2. Right pulmonary veins 3. Thebesian veins *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What vessel runs along the left atrioventricular sulcus?	Circumflex *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
Is valvular function active or passive?	Passive *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What are the structural differences between the right and left ventricles?	The right ventricle is relatively thin-walled compared to the left, and in short-axis section, the right ventricle is crescent shaped. The left ventricle's short axis section appears more egg-shaped, and the wall muscle is about 3 times as thick as the right ventricle due to the extreme force it must generate to contract against systemic vascular resistance. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What structure separates the atria from the ventricles and what is its purpose?	The structure that separates the atria from the ventricles is the cardiac skeleton. It is a fibrous ring that prevents myocardial continuity and interconnects all of the valves. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the fossa ovalis?	It is a depression located in the posterior wall of the interatrial septum that corresponds to the former fetal foramen ovale. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What structure separates the two atria?	The interatrial septum *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What structure separates the two ventricles?	The interventricular septum which serves as a common wall between both ventricles. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*

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What is the significance of the drainage of thebesian veins into the left atrium?	They constitute an anatomical shunt (venous blood entering the arterial circulation) *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What valves close during systole and why?	The tricuspid and mitral valves close during systole to prevent the regurgitation of blood into the atria. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
The right atrium is divided into two components. Name them and describe their anatomical differences.	1. Anterior portion: thin walled 2. Posterior portion: smooth walled *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
Contrast the anatomical shape of the left atrium with the right.	The left atrium is elongated, smooth walled, and smaller than the right atrium. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the appendage on the right atrium called, how is it shaped, and what is its function?	The appendage on the right atrium is the right auricle. It is located on the ventral border of the right atrium, is triangular in shape, and increases the overall surface area of the right atrium. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the name for the muscular ridges located in the ventral aspect of the right atrium?	Trabeculae *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What four structures drain blood into the right atrium?	1. Superior vena cava - 2. Inferior vena cava - 3. Coronary sinus - 4. Thebesian veins *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
About how many liters per day does the heart pump?	7,200 *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
About how many times a day does the heart contract?	100,000 *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
To what degree does atrial contraction contribute to cardiac output?	Normal atrial systole improves cardiac output by about 30%. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*

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What two valves are known as the semilunar valves?	The aortic and pulmonic valves *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What vessel is located along the atrioventricular sulcus?	The right coronary artery *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What regions of the heart do the marginal branches of the RCA provide flow for?	The marginal branches provide blood to the proximal portions of the conduction system, the right atrium, and anterior wall of the right ventricle. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What areas of the heart does the LCA provide flow to?	The LAD with distal septal and diagonal branches supplies the anteroapical portions of the left ventricle, anterior two-thirds of the interventricular septum, and the distal portion of the conduction system. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What areas of the heart does the posterior descending coronary artery provide blood flow for?	The posterior portion of the right and left ventricles, inferior wall of the left ventricle, and the posterior one-third of the interventricular septum. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What structures prevent the retrograde movement of stroke volume?	the semilunar valves *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
From what structure do the chordae tendinae originate?	The papillary muscles. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
Describe the thickness of the interventricular septum.	Most of the interventricular septum is as thick as the left ventricular wall. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What function do the chordae tendinae perform?	They do NOT assist in valve closure. In contrast, they serve to prevent bulging of the leaflets into the atria during systole. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*

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Why is the SA Node the dominant pacemaker in the heart?	The rate at which the resting membrane potential decays during diastole is faster than in other pacemaker cells which provides it with a faster intrinisic rate. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What structure is located along the free edges of the semilunar valves?	tiny fibrous nodules called noduli arantii. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What two valves are known as the atrioventricular valves?	The mitral and tricuspid valves. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
Describe the composition of the valves.	They are composed of fibrous tissue and covered with epithelium. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What are the names of the thin cords attached to the AV valves?	Chordae tendinae *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
Which cusp of the aortic valve is not adjacent to the coronary orifices?	The posterior valve which is known as the noncoronary cusp. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What valves close during diastole?	The pulmonary and aortic semilunar valves. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the function of the cardiac valves?	To prevent retrograde movement of blood into the cardiac chambers. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
Are the AV or semilunar valves more flexible?	The AV valves are more flexible than the rigid semilunar valves. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
Between what two points does the unipolar lead aVF measure voltage?	Between the left foot and the center of the heart. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*

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Between what two points does the unipolar lead aVR measure voltage?	Between the RA and the center of the heart. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
Each of the chest leads are visualized as passing through what anatomic point?	AV node *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
Where do you place each of the six chest leads (V1-V6)?	V1 is placed just right of the sternal border at the 4th intercostal space. V2 is placed just left of the sternal border at the 4th intercostal space. V4 is placed at the midclavicular line 5th intercostal space. V3 is placed midway betwen V2 & V4. V5 is placed at the anterior axillary line horizontal to V4. V6 is placed at the midaxillary line horizontal to V5 *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
Between what two points does the unipolar lead aVL measure voltage?	Between the LA and the center of the heart. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
At what velocity do the internodal pathways conduct impulses from the SA node to the AV node?	0.3 meters per second *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is a normal CK-MB level?	less than 5% relative to the total CK level. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
How long after having an MI is a patient at significant risk for having another cardiac event during surgery?	6 months *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is a problem that is introduced when you change the gain on an ECG monitor and what is the ideal setting it should be on?	Increasing the gain will cause nonsignificant deviations from the baseline to appear significant and vice versa. The best gain setting is 10mm/mV. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
How do you determine whether a Q wave is significant?	The height of the Q wave must be 1/3 the height of the QRS complex or >= 1mm in width. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*

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What R wave changes cana be indicative of ischemic changes?	Significant reduction in R wave amplitude with or without a gain is S wave depth. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
If a patient is at risk for myocardial injury but has a normal 12-lead ECG prior to surgery, which leads which be the best to use for monitoring for ischemia?	V3 and limb lead III *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
How does trauma and surgery affect troponin levels?	It isn't affected at all. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
How can you use 12-lead ECG to differentiate between transmural and nontransmural infarctions?	12-leads are unreliable in making an anatomical distinction between transmural and nontransmural infarctions. Both types of MIs may exhibit Q waves or isolated changes in the ST segment or T wave. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the most accurate method of diagnosing an MI?	Cardiac troponin levels are the most specific test for MI. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
You are performing a 12-lead on a patient who states he had an MI two years ago. Should you expect to see a Q wave on his ECG?	It's probable that he will, but it is important to note that about 14% of patients will lose their Q wave within 3 years. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What electrolyte disturbance would commonly be associated with ST segment changes?	Hypokalemia *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
How long does cardiac troponin stay elevated?	5-10 days *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the problem with using a low bandwidth (0.5-20Hz) filter mode on the ECG monitor?	It can introduce artifact on slower electrical events such as ST and T wave changes that can cause distortion of the readout. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
How long does a CK-MB stay elevated?	48-72 hours *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*

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How soon after the onset of myocardial ischemia will CK-MB and cardiac troponin levels begin to increase?	3-12 hours *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What would cause you to suspect a a right ventricular infarct?	ST segment changes in V1-V4. The greatest specificity is observation of changes in V4R. Q waves are rarely seen in RV infarcts. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What would cause you to suspect an atrial infarction?	Depressed PR segments and alterations in the shape of the P waves, especially in leads 2 and aVF. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What conditions are capable of producing false pathological Q waves?	1. LVH and RVH - 2. Diffuse myocardial disease - 3. Hypertrophic cardiomyopathy - 4. Anterior or posterior hemiblocks - 5. Acute pancreatitis - 6. Pulmonary embolus - 7. Pneumonia - 8. Ventricular pre-excitation *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
Changes in what leads would indicate an antero-septal infarct?	V1-V4 *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is a normal troponin level?	less than or equal to 0.6 ng/mL *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
Some cardiologists state that the ST point distance should be 0.06mm from the J point and some state 0.08mm. What is the key point you should bear in mind regarding this?	That no matter which distance you use, the point of intersection should remain on the ST segment and not go beyond it. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
The T Wave should always be a positive deflection in what leads?	I, II, and V3-V6. It is always inverted in aVR and can be variable in the other leads. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What are the three primary indices of myocardial ischemia on ECG?	1. T Wave changes 2. ST Segment changes 3. Onset of new dysrhythmias *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*

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What T wave changes would you consider to be indicative of ischemia?	T wave peaking and symmetry followed by inversion. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the normal height of the T Wave?	Up to 10mm in the precordial leads and 5mm in the limb leads. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What type of noncardiac surgery places patients at the greatest risk for a cardiac event during surgery?	Aortic surgery *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What T Wave changes would you consider to be normal in a patient with Mitral Valve Prolapse?	Inverted or biphasic T waves in the inferior or anterolateral leads. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is a drawback to using the ECG machine's monitor mode to detect ST changes and what is a way to counteract it?	The monitoring mode reduces baseline wandering using electronic filters, but unfortunately the filters introduce artifact that can be interpreted as ST changes or T wave inversion. To prevent this, you can place the monitor in diagnostic mode which eliminates potential false ST and T wave changes. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What are the three ST changes that indicate ischemia?	1. ST depression >= 1mm 2. ST upsloping or downsloping >= 1mm measured 0.08 secs from the J point 3. ST elevation >= 1mm *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
Is a change in the ST segment or T wave considered to indicate a greater indicator of ischemia?	ST segment changes reflect a more severe degree of ischemia. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
How do you use the J point to determine the degree of ST elevation or depression?	You draw a horizontal line 0.08 seconds (2mm) from the J point and then draw a vertical line from that point that intersects with the ST segment. The point at which the lines intersect indicates the distance from the isoelectric line (which is the mm of elevation or depression). *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What three items should be verified when using automatic ST segment analysis?	You should verify accurate positioning of the 1) iso point, 2) J point, and 3) ST point. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*

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Electrocardiogram

What percentage of patients with CAD but not experiencing acute cardiac pain will present with normal ECGs?	50-70% *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is pseudonormalization of the T Wave?	When a patient has a baseline inverted T Wave which becomes upright in response to an ischemic episode. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the J point and what purpose does it serve?	It is the junction between the S wave and the ST segment. It allows the clinician to determine the magnitude of nonhorizontal (upsloping or downsloping) ST segment changes. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
How can patient positioning affect the ST segment?	The prone position can cause ST changes *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is a neurologic injury that can cause ST changes?	Subarachnoid hemorrhage *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What are three gastrointestinal diseases associated with ST changes?	Acute abdomen, hiatal hernia, and gallbladder disease *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What ST segment changes would you assume to be significant in pediatric patients?	ST deviations >= 1-2mm in the limb leads or up to 3mm in the right precordial leads can be due to digoxin, myocardial disease, or pericardial disease. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What reciprocal changes would you expect to see if you noticed Q waves, ST segment changes, or T wave changes in the anterior leads (V1-V5, aVL)?	You would see reciprocal changes in the inferior leads (II,III, & aVF). For example, Q waves in the anterior leads would manifest as heightened R waves in the inferior leads. ST segment elevation would be seen as ST segment depression, and inverted T waves would be seen as tall, peaked T waves in the reciprocal leads. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
How would you interpret ST segment deviations in an adolescent or younger patient?	It can be a normal occurrence, particularly if previous ECGs demonstrated the same occurrence. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*

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Although ST segment elevation is a more ominous sign than ST segment depression, what is a concern when analyzing ST segment depression in a single lead?	It could be a reciprocal change to a ST segment elevation in another lead. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What are five drugs that can cause ST changes?	Diuretics, digoxin, guanethidine, quinidine, and insulin *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What symptoms would you expect to see with supraventricular tachycardia?	Supraventricular tachycardia (SVT) is commonly associated with lightheadedness, dizziness, syncope, fatigue, and chest discomfort. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 77.*
What are 8 common causes of atrial flutter?	1. Reentrant phenomenon 2. COPD with cor pulmonale 3. Pulmonary embolism 4. Valvular disease 5. CAD or acute MI 6. Hyperthyroidism 7. Electrolyte abnormalities 8. Toxic effects of drugs (digoxin, quinidine) *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
Is atrial fibrillation or atrial flutter considered more unstable?	Atrial flutter is more unstable. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
When should you consider anticoagulating a patient with atrial fibrillation before cardioversion?	If the atrial fibrillation has been present for more than two weeks. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What are seven common causes of atrial tachycardia?	1. Hypovolemia 2. Hypotension 3. CHF 4. Fever 5. Malignant hyperthermia 6. Inadequate analgesia 7. WPW syndrome *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
At what heart rate is atrial fibrillation considered to be "uncontrolled"?	Greater than 130 bpm. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
How do you diagnose Wolff-Parkinson-White syndrome?	By the presence of a short PR interval and a delta wave (slurred upstroke of the QRS complex) *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*

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What is the difference between SVT and atrial tachycardia?	Supraventricular tachycardia is any tachycardia originating above the ventricles, thus, sinus tach, junctional tach, and atrial tach are all forms of SVT. SVT most commonly occurs due to a reentry circuit consisting of anterograde conduction over the slower AV nodal pathway and retrograde conduction over a faster accessory pathway. It can also occur due to enhanced automaticity of secondary pacemaker cells. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 77-79.*
If you are checking the pulse of a patient with atrial fibrillation and find that the pulse is regular, what two events could be responsible for the regular rhythm?	1. Spontaneous conversion to NSR 2. Complete AV block (as can occur sometimes with the use of digoxin) and the development of a junctional escape rhythm. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
How is the visibility of the P waves and T waves affected by the high rates associated with atrial tachycardia?	The P waves can be hidden in the preceding T waves, but also the T waves can appear hidden and only the P wave and QRS complexes are visible. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What pharmacotherapy can increase the success rate of cardioversion in atrial fibrillation?	The use of quinidine, procainamide, or norpace 48 hours prior to cardioversion can increase the success rate. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the most common arrhythmia in pediatric patients undergoing anesthesia?	Bradydysrhythmias *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the heart rate range for atrial tachycardia?	150-250 *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
Where is the origin of the electrical impulses during atrial tachycardia?	The atrial musculature *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What heart rate defines sinus bradycardia?	less than 60 *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*

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The diagnosis of normal sinus rhythm is based upon what finding?	The SA rate (p wave) is between 60-100 beats per minute. Technically, QRS's do not have to be present to diagnose NSR (that would be called NSR with ventricular asystole). For example, you could have NSR with first degree AV block as long as the p wave rate is between 60-100 beats per minute. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
When determining heart rate by the R-R' distance, you can guage the rate based on 5mm intervals. What is the sequence of rates according to 5mm intervals?	300, 150, 100, 75, 60, 50, 43, 37, 33. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the heart rate range for normal sinus rhythm?	60-100 *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
An arrhythmia due to an alteration in automaticity would be due to what phase of the slow response action potential?	Phase 4 *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
An arrhythmia due to a change in conductivity would be due to a change in what phase of the action potential?	Phase 0 *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What are the two possible causes of a wide QRS complex?	The impulse originated in the ventricle, or it was supraventricular with aberrant conduction. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
How does the P wave appear in lead 2 if atrial depolarization originates from the sinus node? The AV Node?	If it originates from the sinus node, the P wave will be upright in lead 2. If it originates from the AV node, it will be inverted in lead 2. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the most common arrhythmia in unanesthetized pediatric patients?	SVT *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What leads should you use to distinguish atrial tachycardia with aberrant ventricular conduction from ventricular tachycardia?	Leads V1, V6, MCL1, and MCL6 are best for assessing P waves in atrial tachycardia. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*

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What are the four most common arrhythmias in adult patients undergoing anesthesia?	1. PACs 2. Bradycardias 3. Nodal rhythms 4. Simple PVCs *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
How would you expect anticholinesterase drugs to affect heart rate?	They can cause bradycardia. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the onset time, peak effect time, and half-life of digoxin?	Onset: 5-30 minutes. Peak: 1.5-5 hours. Half-life: 36 hours *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
When do the majority of perioperative arrhythmias occur?	72% occur during tracheal intubation and extubation. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the therapeutic range for digoxin?	0.5-2 ng/mL *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the incidence of perioperative arrhythmias in patients undergoing cardiac and noncardiac surgery?	16.3 to 84% *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What types of arrhythmias can result from WPW syndrome?	Any type of premature beat, SVT, atrial flutter, and atrial fibrillation. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the mortality rate associated with digoxin induced VTach?	50% *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What medical conditions are associated with WPW syndrome?	1. VSD 2. MVP 3. Transposition of the great vessels 4. Ebstein's anomaly (abnormal insertion of the tricuspid valves which leads to an atrialized right ventricle) *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
Other than digitalis, diltiazem, and verapamil, what drugs can cause problems in the treatment of tachycardia in patients with WPW?	Lidocaine, propranolol, and adenosine. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*

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What can result from the administration of digitalis, verapamil, and diltiazem in patients with WPW?	Although they effectively suppress transmission of impulses through the AV node, they are less predictable in inhibiting conduction through accessory tracts, and a further increase of the HR can occur as a result. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the typical treatment of patients with WPW associated tachycardia?	Although quinidine and disopyramide may be able to control the HR, surgical ablation of the accessory tracts is usually required. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What does the delta wave in WPW syndrome represent?	Premature septal depolarization via an accessory tract. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What are the risk factors that predispose patients to digoxin toxicity?	1. Hypokalemia 2. Hypercalcemia 3. Renal impairment 4. Pulmonary disease (hypoxemia) 5. Drugs known to alter the pharmacokinetics of digoxin such as verpamil, nifedipine, esmolol, and quinidine. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the IV infusion dosing regimen for epinephrine in the treatment of sinus bradycardia?	2-10 mcg/min IV *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is responsible for the difference between conducted, nonconducted, and aberrantly conducted PACs?	Conducted PACs produce a QRS after the premature P wave. Nonconducted P waves arrive at the AV node while it is still refractory, so the QRS does not appear. Abberantly conducted PACs can appear in a few different ways. If the P wave hits the AV node during its relative refractory period, the beat can occur with a first degree block. If a portion of the distal conducting system is refractory, then a wide QRS could appear due to a bundle branch block. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is a potential complication if a PAC occurs during the T wave?	It could result in atrial fibrillation or atrial flutter. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
How do you diagnose sick sinus syndrome?	Carotid massage will generate a period of sinus arrest longer than 3 seconds. A valsalva maneuver will generate only a slight reduction in heart rate. Patients with SSS will also have a reduced response to atropine. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*

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What are four possible causes of PACs?	1. Anxiety and stimulants 2. Elevated atrial pressure 3. Ischemia, injury, and infarction 4. Digoxin toxicity *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
How do you treat symptomatic bradycardia in patients with sick sinus syndrome?	Atropine (may require a larger dose than expected) and isoproterenol (2-20 mcg/min) *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the underlying cause of sick sinus syndrome and what type of patients typically have it?	It is a deficiency of normal subsidiary pacemakers (even though the patient is capable of SVT). Patients with SSS are generally older and have a history of CAD. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What are the three groups of rhythm disturbances associated with sick sinus syndrome?	1. Chronic bradycardia 2. Intermittent sinus arrest with various escape rhythms 3. Tachy-Brady syndrome where the patient exhibits alternating bouts of SVT and bradycardia *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is a nonpharmacologic technique for treating bradycardia?	transcutaneous or transvenous pacing *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
How do you differentiate between sinus pause, sinus exit block, and sinus arrest?	Sinus pause: the period of electrical silence lasts less than one sinus cycle. Sinus exit block: exactly two cycles will elapse during the electrical silence. Sinus arrest: more than two cycles will be skipped. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the IV infusion dosing regimen for isoproterenol in the treatment of sinus bradycardia?	2-10 mcg/min IV (same as epinephrine) *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the heart rate range for atrial flutter (atrial rate)?	250-350 *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
Periods of electrical silence between cardiac cycles are identified as what three possible entities?	Sinus pause, sinus exit block, or sinus arrest. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*

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What is the heart rate range for sinus tachycardia?	100-150 *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
In what particular class of patient will atropine be ineffective for the treatment of bradycardia?	Patients who have previously had a cardiac transplant *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
How do you diagnose sinus dysrhythmia?	By a 1-1 conduction originating in the SA node with an irregular R-R interval. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the term for an SA dysrhythmia that is associated with a physiologic event (such as the respiratory cycle)?	Phasic sinus rhythm *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
How would you expect succinylcholine to affect cardiac rhythm?	Repeat doses of succinylcholine can cause sinus bradycardia. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the atrial rate range for atrial fibrillation?	greater than 350 *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the term for an SA dysrhythmia that is not associated with a physiologic event such as the respiratory cycle?	It is simply called a sinus irregularity. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the dosing regimen for ephedrine in the treatment of sinus bradycardia?	5-20mg IV *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the dose of glycopyrrolate for the treatment of sinus bradycardia?	0.2mg IV (may repeat 0.2mg doses prn) *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
What is the IV infusion dosing regimen for dopamine in the treatment of sinus bradycardia?	3-7.5 mcg/kg/min IV *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*

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What is the dosing regimen for atropine for the treatment of sinus bradycardia?	0.4 - 1mg IV (may repeat dose Q 5min to a total dose of 0.04 mg/kg). Increase the dose by 2-2.5 times if administered via ETT. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
In what setting is atropine contraindicated for the treatment of sinus bradycardia and why?	In the setting of an acute MI because it can lead to VT of VF. *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
Name 6 drugs you could use intraoperatively to treat bradycardia.	1. Glycopyrrolate 2. Atropine 3. Ephedrine 4. Epinephrine 5. Dopamine 6. Isoproterenol *Kossick MA. EKG Interpretation: Simple-Thorough-Practical. 2nd ed. Parkridge, IL: AANA Publishing; 2000.*
How is polyuria related to supraventricular tachycardia?	Polyuria may result from SVT (or any other atrial tachycardia resulting in AV dys-synchrony) due to an increased secretion of atrial natriuretic peptide due to the increased atrial pressures that result from atrial contraction against a closed AV valve. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 77.*
What symptoms are associated with supraventricular tachycardia (SVT)?	Supraventricular tachycardia (SVT) is commonly associated with lightheadedness, dizziness, syncope, fatigue, and chest discomfort. It most commonly occurs due to a reentry circuit consisting of anterograde conduction over the slower AV nodal pathway and retrograde conduction over a faster accessory pathway. It can also occur due to enhanced automaticity of secondary pacemaker cells. Polyuria may result from SVT (or any other atrial tachycardia resulting in AV dys-synchrony) due to an increased secretion of atrial natriuretic peptide due to the increased atrial pressures that result from atrial contraction against a closed AV valve. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 77.*
SA node depolarization is transmitted from the right atrium to the left atrium via	Bachmann's bundle, which is a branch of the anterior internodal pathway. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 243.*
What are the characteristics of a left bundle branch block?	The characteristics consistent with a left bundle branch block are: QRS > 0.12 secs, lack of septal Q waves in V4-V6, I and aVL, RR' QRS pattern in I, aVL, and V4-V6, and secondary ST or T wave changes in I, aVL, and V4-V6. A deep, rounded S wave in leads I and aVL is typical of a right bundle branch block while a V5 amplitude > 26 mm is one of the characteristics of left ventricular hypertrophy. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1704.*

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Electrocardiogram

What leads on a 12-lead ECG monitor the lateral portion of the heart? Inferior? Anterior?

The lateral aspect of the heart is best evaluated using leads I, aVL, V5, and V6. The inferior aspect is best monitored by leads II, III, and aVF, and the anterior portion of the heart is best assessed using leads V1-V4.

Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1705-1709.

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